Implement a CNN for character recognition using the EMNIST dataset

To implement a Convolutional Neural Network for handwritten character recognition using the EMNIST Balanced dataset.

Algorithm
1. Import TensorFlow and TensorFlow Datasets libraries.
2. Load the EMNIST Balanced dataset.
3. Divide the dataset into training and testing portions.
4. Normalize the image pixel values to the range 0–1.
5. Correct the orientation of the EMNIST images.
6. Shuffle the training dataset and divide it into batches.

7. Create a CNN model with convolutional layers.
8. Add max-pooling layers to reduce image dimensions.
9. Add a flatten layer to convert feature maps into a vector.
10. Add a dropout layer to reduce overfitting.
11. Add a softmax output layer with 47 neurons representing the 47 classes.
12. Train and evaluate the model using the prepared datasets.

Program

```
import tensorflow as tf
import tensorflow_datasets as tfds
from tensorflow.keras import layers, models
tf.keras.utils.set_random_seed(12)
# Load EMNIST Balanced dataset
(ds_train, ds_test), info = tfds.load(
'emnist/balanced',
split=['train[:80%]', 'test[:20%]'],
as_supervised=True,
with_info=True
)
# Preprocess images
def prepare(image, label):
image = tf.cast(image, tf.float32) / 255.0
image = tf.transpose(image, [1, 0, 2])
return image, label
train = (
ds_train.map(prepare)
.shuffle(10000)
.batch(128)
.prefetch(tf.data.AUTOTUNE)
)
test = (
ds_test.map(prepare)
.batch(128)
.prefetch(tf.data.AUTOTUNE)
)
# Create CNN
model = models.Sequential([
layers.Input(shape=(28, 28, 1)),
layers.Conv2D(32, 3, activation='relu', padding='same'),
layers.MaxPooling2D(),

layers.Conv2D(64, 3, activation='relu', padding='same'),
layers.MaxPooling2D(),
layers.Flatten(),
layers.Dropout(0.3),
layers.Dense(128, activation='relu'),
layers.Dense(47, activation='softmax')
])
# Compile model
model.compile(
optimizer='adam',
loss='sparse_categorical_crossentropy',
metrics=['accuracy']
)
# Train model
model.fit(
```

```
train,
epochs=3,
validation_data=test
)
# Evaluate
loss, accuracy = model.evaluate(test, verbose=0)
print("EMNIST Test Accuracy:", accuracy * 100, "%")
```

```
Output of Program
Downloading EMNIST Balanced dataset...
Epoch 1/3
Epoch 2/3
Epoch 3/3
EMNIST Test Accuracy: Training-dependent value
```

Result

Thus, a CNN model was successfully implemented for handwritten character recognition using the EMNIST Balanced dataset. Normalization, convolution, pooling and dropout techniques were applied to improve learning.